Notes on Computational Fluid Dynamics: General Principles

2.14 Diffusion

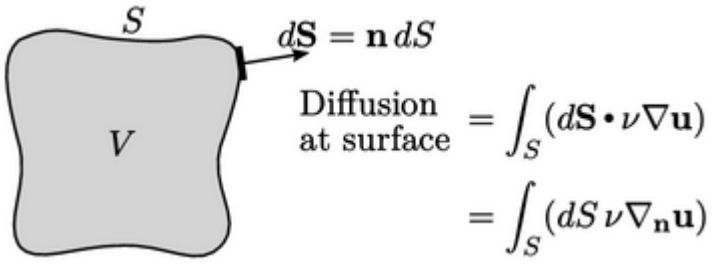
The momentum equation for a homogeneous, incompressible, Newtonian fluid is presented in Eq. (2.47). In the case of zero body force, $\mathbf{b} = \mathbf{0}$, and $\nu =$ constant, Eq. (2.47) becomes

$$\frac{\partial \mathbf{u}}{\partial t} + \nabla \bullet (\mathbf{u}\mathbf{u}) - \nabla \bullet (\nu \nabla \mathbf{u}) = -\nabla p. \tag{2.49}$$

This is a typical *transport equation* characterised by:

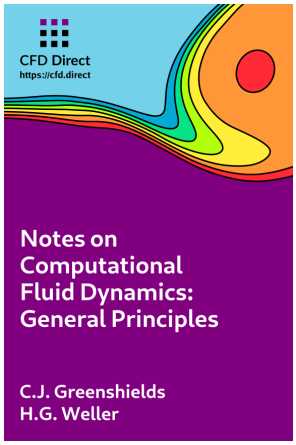
- the local rate of change $\partial \mathbf{u}/\partial t$, described in Sec. 2.5;
- advection of $\mathbf{u}$ by $\nabla \bullet (\mathbf{u}\mathbf{u})$, described in Sec. 2.8;
- diffusion of $\mathbf{u}$ by $\nabla \bullet (\nu \nabla \mathbf{u})$;
- a "source" due to ∇p, described in Sec. 2.12.

Note that, since $\nu =$ constant, $\nabla \bullet (\nu \nabla \mathbf{u}) = \nu \nabla \bullet (\nabla \mathbf{u}) = \nu \nabla^2 \mathbf{u}$, where $\nabla^2 \equiv \nabla \bullet \nabla$ denotes the Laplace operator.



The $\nabla \bullet (\nu \nabla \mathbf{u})$ term is a special form of divergence in which the flux includes the *surface normal gradient* denoted by the operator $\nabla_\mathbf{n}$ where $\nabla_\mathbf{n} \Psi \equiv \mathbf{n} \bullet \nabla \Psi$.

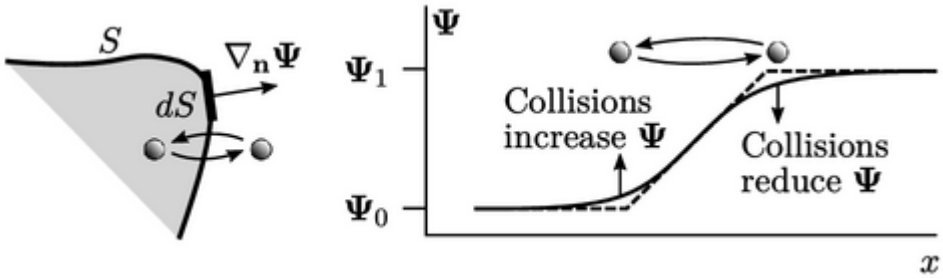
The term models *diffusion* across the surface of the fluid element. Diffusion generally represents the transport of a fluid property — here, momentum — due to *fluctuating motions* that are not captured by the bulk motion that is represented by the *continuum* velocity $\mathbf{u}$.



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Fluctuations include any random motion of particle constituents of matter, *e.g.* molecules, and turbulent structures. Through these motions, particles can pass across a surface boundary, transporting property Ψ through a gradient of Ψ (above, left).

Particles carrying higher Ψ move into regions of particles with lower Ψ and vice versa. Through particle collisions, high values of Ψ tend to reduce and low values increase (right).

Laplacian

"Laplacian"[11] describes a term of the form $\nabla \bullet (\Gamma \nabla \Psi)$ where Γ is a diffusivity coefficient.[12] A Laplacian term is *conservative* since all variables are to the right of a divergence, as described in Sec. 2.9. It is also *bounded* since it tends to decrease high values and increase low values as shown above.

The Laplacian represents a flux due to $\Gamma \nabla_n \Psi$ across the surface, per unit volume, as $\Delta V \to 0$

$$\nabla \bullet (\Gamma \nabla \Psi) = \lim_{\Delta V \to 0} \frac{1}{\Delta V} \int_S (dS\ \Gamma \nabla_n \Psi). \tag{2.50}$$

[11] After Pierre-Simon Laplace, *Théorie des attractions des sphéroïdes et de la figure des planètes*, 1785 but described in 1761 both by Leonhard Euler and Jean-Baptiste le Rond d'Alembert.

[12] More precisely, "Laplacian" denotes a term without Γ, *i.e.* $\nabla \bullet (\nabla \Psi)$, represented as $\nabla^2 \Psi$ but we include Γ here because it is generally needed.

Chris Greenshields Henry Weller 11th April 2022 Notes on CFD

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